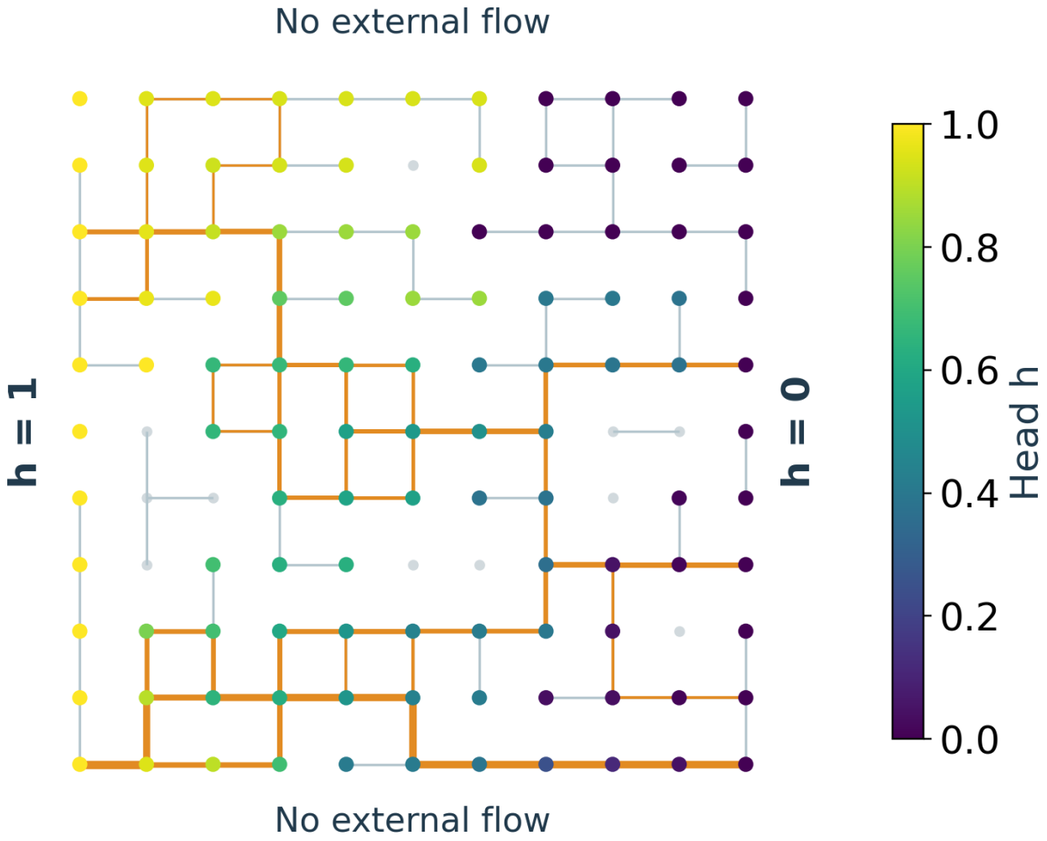
Flow through a random network



A simplified fracture network

25 × 25 nodes; 1,200 potential links.

A fixed driving force

Left head = 1; right head = 0. No external flow at top or bottom.

What do we want to predict?

Total flow and effective conductance: G_eff = Q_total / Δh.

Dimensionless model • Illustration uses 11 × 11 nodes for clarity.

From uncertain links to a flow prediction

01 Connectivity

One global p per network:
$p \sim \text{Uniform}(0.5 - \delta, 0.5 + \delta)$.
Each link is active with probability p.

02 Link conductance

Active links: $\ln(g) \sim \text{Normal}(0, \sigma^2)$.
Inactive links: $g = 0$.
Larger σ means greater heterogeneity.

03 Solve node heads

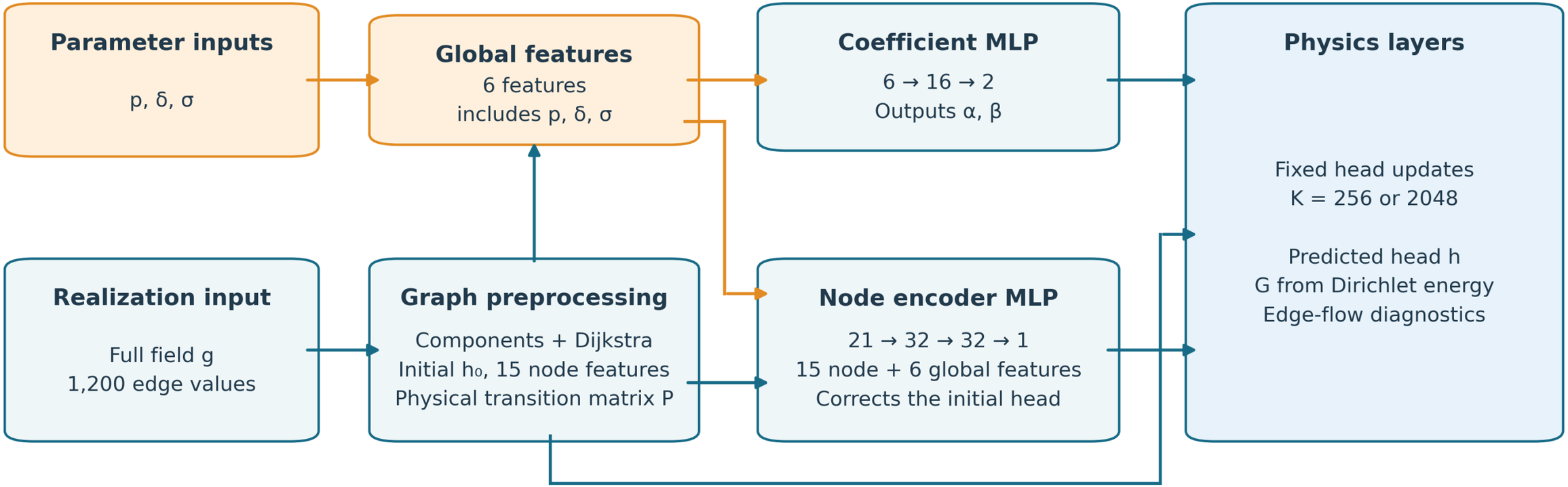
Find spanning components.
Enforce flow balance at free nodes:
$\Sigma_j g_{ij}(h_i - h_j) = 0$.

04 Compute the output

Solve the sparse linear system for all unknown heads together.
Then calculate link flows and G_eff.

δ controls uncertainty in connectivity; σ controls spread in log-conductance.

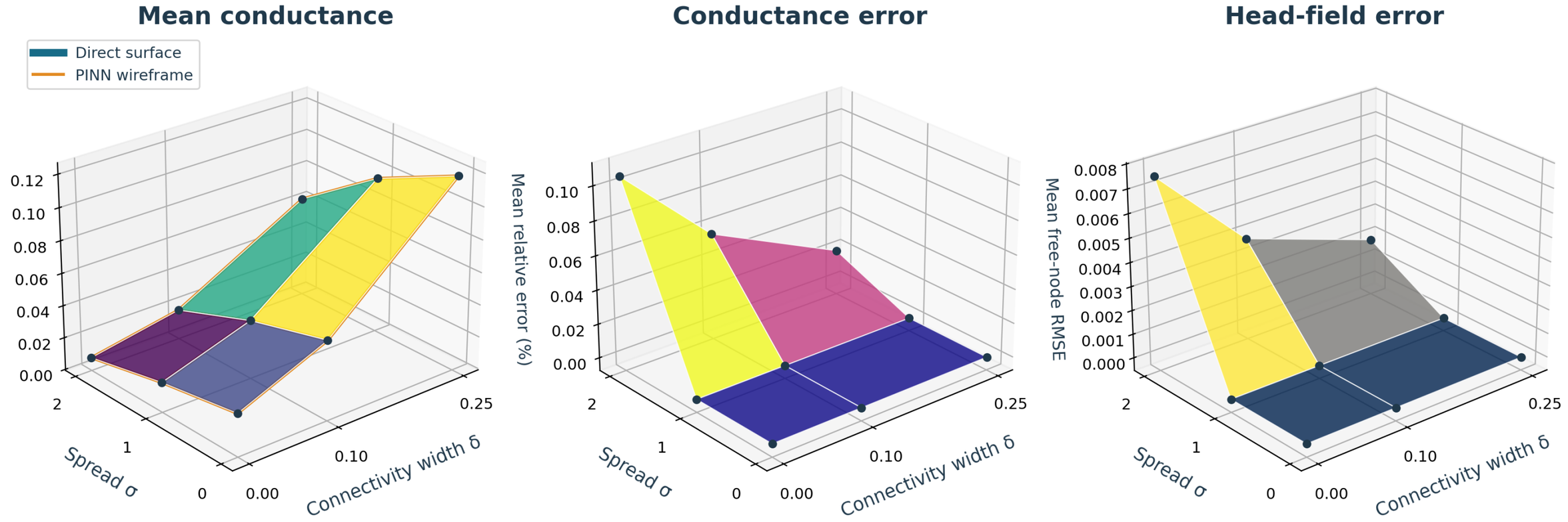
One trained parametric graph PINN



Training: energy + conservation residual loss • 1,939 trainable parameters

Same weights for new fields • Per-input graph preprocessing and K fixed head updates still run • No new sparse solve.

How the response changes across δ and σ



Dots: 9 evaluated parameter pairs. Facets only join those points. Error panels use positive-flow cases.

Does the numerical solution obey physics?

Internal conservation

Maximum node residual

2.75×10^{-13}

$\Sigma_j Q_{ij} \approx 0$

Inlet–outlet balance

Maximum relative mismatch

7.91×10^{-12}

$Q_in \approx Q_out$

Energy consistency

Maximum absolute discrepancy

2.41×10^{-13}

$\Sigma_e g_e(\Delta h_e)^2 \approx Q_in \times \Delta h$

Head bounds

$0 \leq h \leq 1$, within roundoff.

Maximum excess above 1:

4.66×10^{-14}

1,000 independent random networks • float64 • Floating, undefined heads excluded from bounds.

Does the solver reproduce known answers?

Uniform grid

Exact: G = g n / (n − 1).
Linear head profile also verified.
4 sizes × 3 conductances.

Series and parallel paths

Series: G = 1 / Σ(1/g).
Parallel: branch conductances add.
Both agree with analytical answers.

Disconnected network

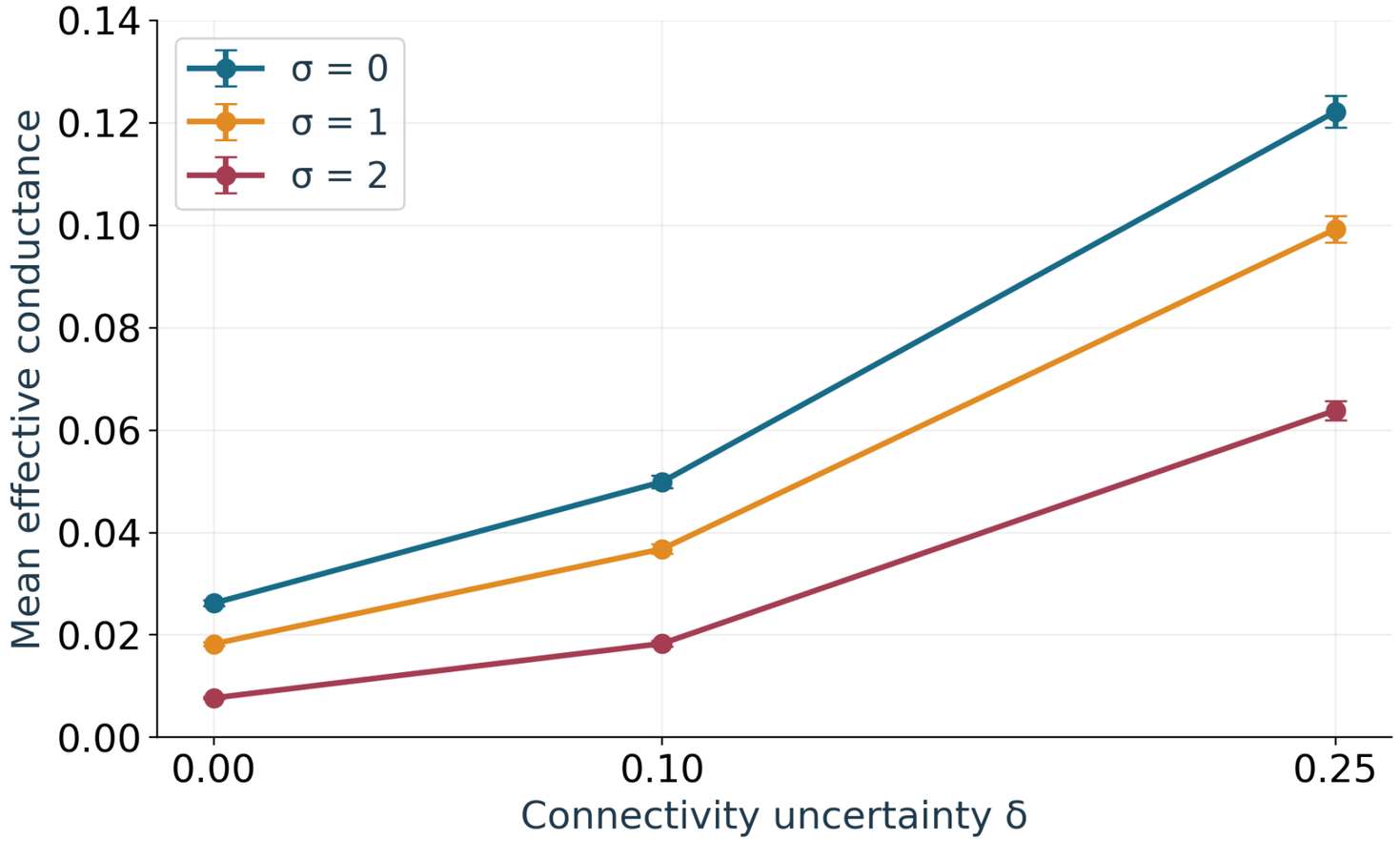
A complete vertical cut removes
every spanning path.
Computed conductance: exactly zero.

Scaling and monotonicity

G(3g) = 3G(g).
Increasing edge conductances
never reduced G in 12 test cases.

39 recorded benchmark / invariant checks: all passed • Original automated suite: 24 tests.

How do uncertainties change the mean flow?



Wider connectivity uncertainty

At σ = 1, mean G increases 0.0183 → 0.0992 as δ: 0 → 0.25.

More heterogeneous conductance

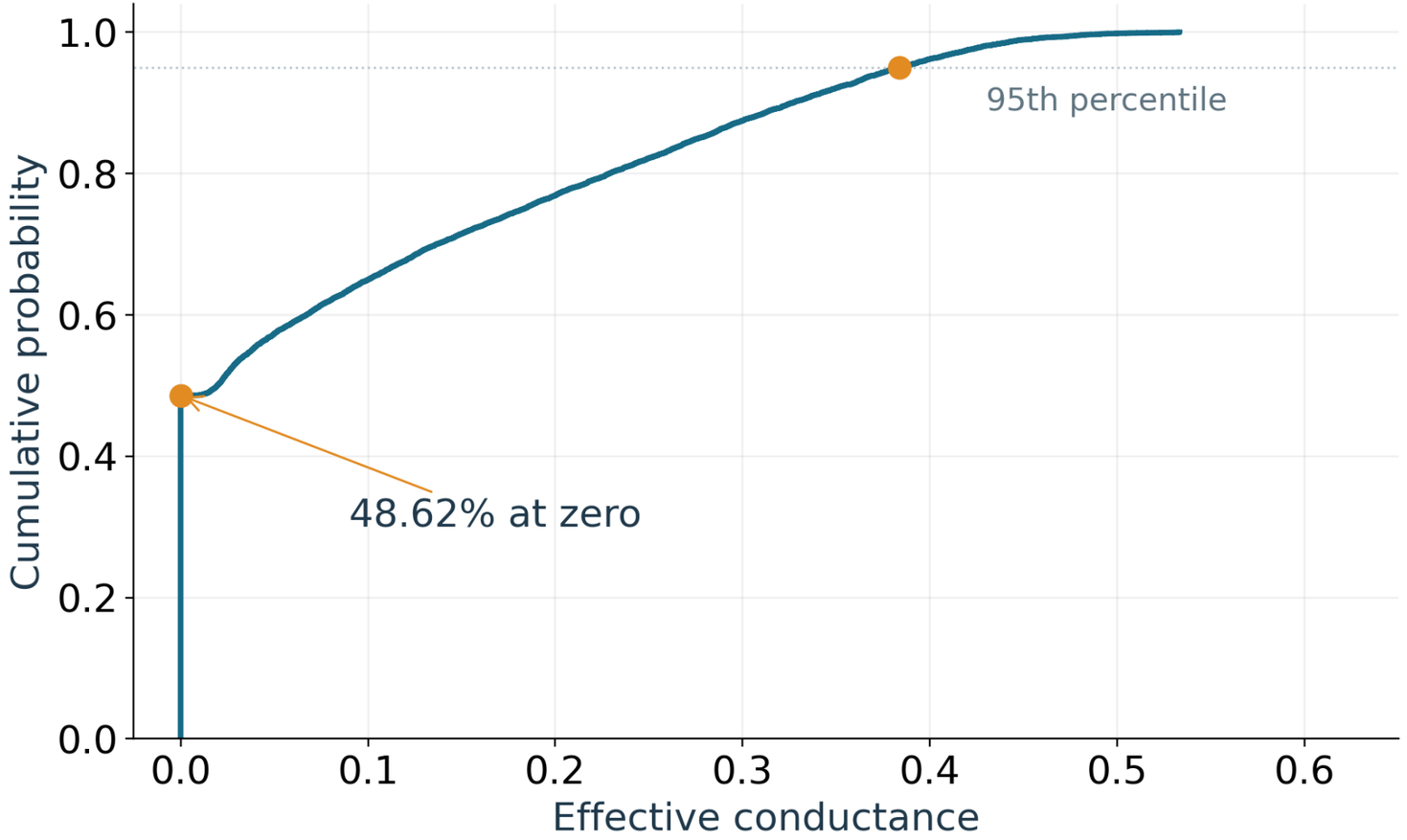
At fixed δ, mean G decreases as σ increases in these scenarios.

Why this happens

Connectivity has a nonlinear response. Weak links can become bottlenecks.

9 scenarios × 10,000 realizations • Error bars: Monte Carlo 95% confidence intervals.

The mean alone hides the output distribution



Almost half have no flow

48.62% of realizations have G = 0.

A broad positive component

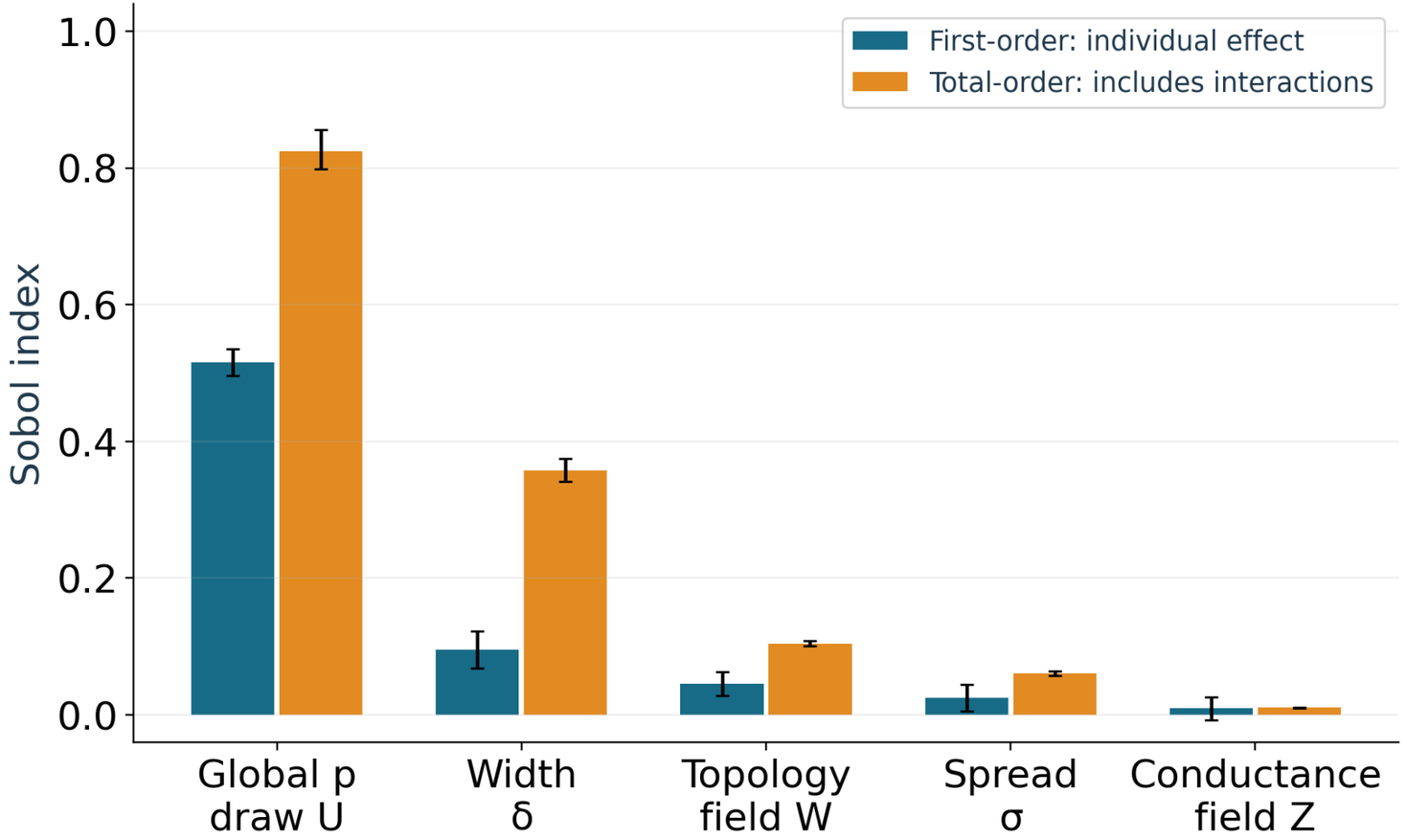
Mean = 0.0992
Median = 0.0195
95th percentile = 0.3838

Report both parts

Connection probability and positive-flow magnitude answer different questions.

Reference scenario: δ = 0.25, σ = 1 • A mixed distribution: an atom at zero plus positive values.

Connectivity is the dominant uncertain influence



Method: grouped Sobol / Jansen

Global variance-based analysis.
First-order: individual effect.
Total-order: includes interactions.

Connectivity dominates

Global p draw U: total index 0.825.
δ: first 0.096, total 0.358.

Interactions matter

δ has a much larger total effect.
Total indices overlap; do not add them.

p = 0.5 + δU; independent groups δ, σ, U, W, Z • 16,384 base pairs • 1,000 bootstrap resamples.

What we learned — and how to extend the model

01 Main finding: connectivity

Connection structure and its interactions dominate output variability over the selected input ranges.

02 Main finding: distribution

Mean conductance is not enough. Report zero-flow probability, conditional flow and an upper quantile.

03 Model limitations

Regular 2D geometry; steady linear flow.
Exploratory priors; finite-size effects.
No geological field calibration.

04 Toward realistic fractured rocks

Use 3D discrete fracture networks, aperture-based conductance and correlation.
Calibrate with head, flow or tracer data.

Numerical verification establishes implementation correctness; it does not establish field prediction accuracy.